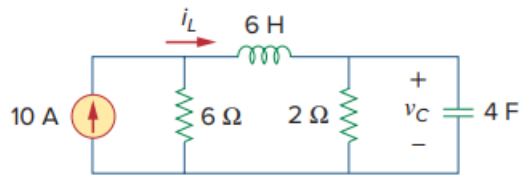


## Problem

Determine  $v_C$ ,  $i_L$ , and the energy stored in the capacitor and inductor in the circuit of Fig. 6.28 under dc conditions.

Fig. 6.28:



## Step-by-step solution

## Step 1 of 4

Consider the following circuit:

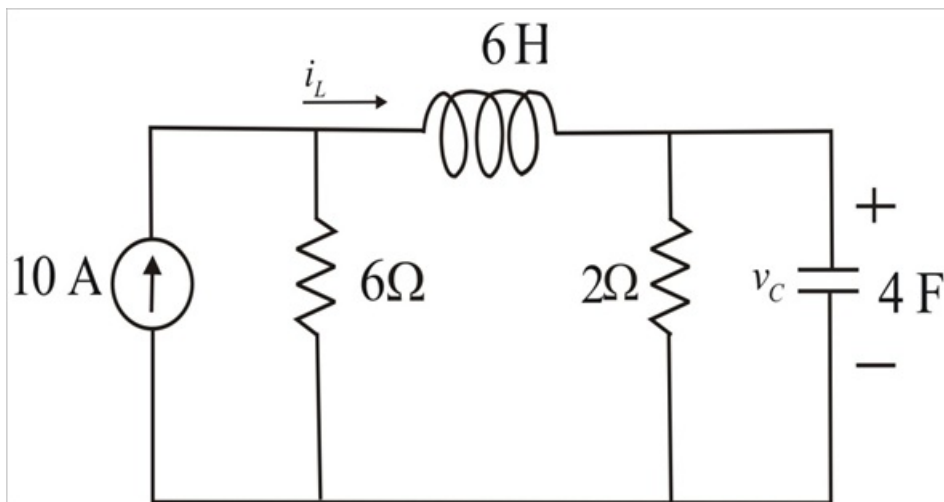


Figure 1

## Step 2 of 4

Under dc conditions, capacitor acts as an open circuit and inductor acts as a short circuit. Redraw the circuit by replacing inductor with a short circuit and capacitor with an open circuit.

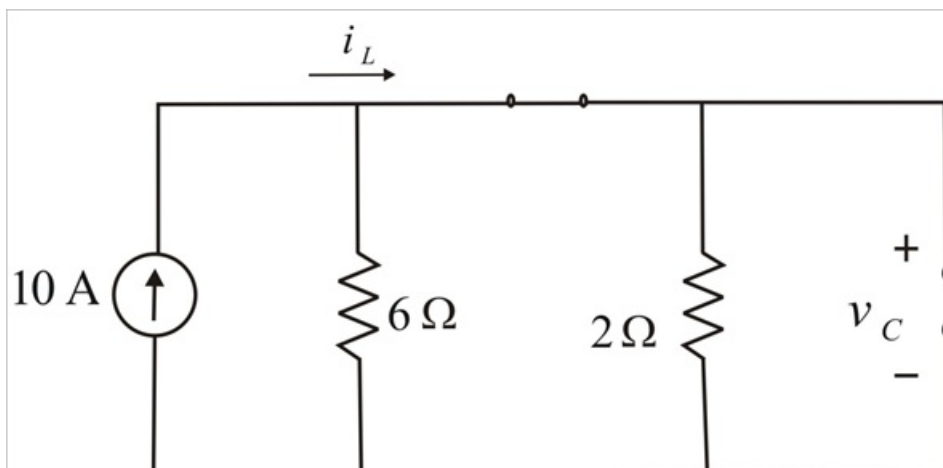


Figure 2

**Step 3** of 4

Determine current  $i_L$  flowing through  $2\ \Omega$  resistor by using current division rule.

$$\begin{aligned} i_{2\Omega} &= \left( \frac{6}{6+2} \right) (10) \\ &= \left( \frac{6}{8} \right) (10) \\ &= \frac{15}{2} \\ &= 7.5\text{ A} \end{aligned}$$

Here, current through the inductor is  $i_L$ , which is same as the current through  $2\ \Omega$  resistor.

Hence, the current through the inductor is,

$$i_L = i_{2\Omega} = 7.5\text{ A}$$

Thus, the current through the inductor,  $i_L$  is **7.5 A**.

Determine the voltage across the capacitor.

$$\begin{aligned} v_C &= 2i_L \\ &= (2)(7.5) \\ &= 15\text{ V} \end{aligned}$$

Thus, the voltage across the capacitor,  $v_C$  is **15 V**.

**Step 4** of 4

Determine the energy stored in the  $4\text{ F}$  capacitor.

$$\begin{aligned} w_C &= \frac{1}{2} C v_C^2 \\ &= \left( \frac{1}{2} \right) (4) (15)^2 \\ &= 450\text{ J} \end{aligned}$$

Hence, the energy stored in the capacitor is **450 J**.

Determine the energy stored in the  $6\text{ H}$  inductor.

$$\begin{aligned} w_L &= \frac{1}{2} L i_L^2 \\ &= \left( \frac{1}{2} \right) (6) (7.5)^2 \\ &= 168.75\text{ J} \end{aligned}$$

Thus, the energy stored in the inductor is **168.75 J**.